

Original Research

Learning to match patients to clinical trials using large language models

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ABSTRACT

Objective: This study investigates the use of Large Language Models (LLMs) for matching patients to clinical trials (CTs) within an information retrieval pipeline. Our objective is to enhance the process of patient-trial matching by leveraging the semantic processing capabilities of LLMs, thereby improving the effectiveness of patient recruitment for clinical trials.

Methods: We employed a multi-stage retrieval pipeline integrating various methodologies, including BM25 and Transformer-based rankers, along with LLM-based methods. Our primary datasets were the TREC Clinical Trials 2021–23 track collections. We compared LLM-based approaches, focusing on methods that leverage LLMs in query formulation, filtering, relevance ranking, and re-ranking of CTs.

Results: Our results indicate that LLM-based systems, particularly those involving re-ranking with a fine-tuned LLM, outperform traditional methods in terms of nDCG and Precision measures. The study demonstrates that fine-tuning LLMs enhances their ability to find eligible trials. Moreover, our LLM-based approach is competitive with state-of-the-art systems in the TREC challenges.

The study shows the effectiveness of LLMs in CT matching, highlighting their potential in handling complex semantic analysis and improving patient-trial matching. However, the use of LLMs increases the computational cost and reduces efficiency. We provide a detailed analysis of effectiveness-efficiency trade-offs.

Conclusion: This research demonstrates the promising role of LLMs in enhancing the patient-to-clinical trial matching process, offering a significant advancement in the automation of patient recruitment. Future work should explore optimising the balance between computational cost and retrieval effectiveness in practical applications.

1. Introduction

The advancement of medical science, especially in the innovation of novel treatments and medical technologies, is largely due to learnings derived from clinical trials (CTs) [1]. However, a significant barrier to the success of these trials is the recruitment of patients, a process hampered by limited awareness and accessibility for both patients and physicians [1,2]. The recruitment phase is laborious, entailing the meticulous screening of candidates for eligibility [3]. The automation of this process has the dual advantage of expediting translational science [2] and facilitating patient recruitment in appropriate CTs [4].

Automated matching systems are proposed to streamline the matching of patients with suitable CTs. The problem is often modelled and approached as an information retrieval task within the patient-trials evaluation framework [4–7]. This process involves complex semantic analysis to align patient data with the multifaceted clinical trial documents [8,9]. This complexity is further amplified by the intricate

language and medical terminologies used in the eligibility criteria, often presented in either semi-structured or unstructured manner [10].

With the emergence of *Pretrained Language Models* (PLMs), there have been studies to utilise their medical language processing capability to rank the relevance of clinical trials to a given patient [8,9]. Other approaches introduce the automation of the process of rewriting medical text to optimise matching patients to clinical trials [11].

The latest PLMs—generative *Large Language Models* (LLMs)—such as ChatGPT, have been applied to tasks such as diagnosis prediction and formulation of systematic review queries [12]. However, these models have often been critiqued for their lack of tractability and inability to consistently produce desirable citations [13,14].

In this study, we investigate the use of LLMs for matching patient profiles to clinical trials within an information retrieval pipeline. In such a setup, LLMs are used to improve the process of returning relevant documents from a specific corpus. Specifically, we focus on

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a system that takes a patient description as input and outputs a list of CTs ordered by relevance.

We introduce a broad selection of methods that leverage the LLMs at critical stages of an information retrieval pipeline, including query formulation and relevance ranking. We compare the LLM-based approaches against a backdrop of established techniques (including simpler or smaller PLMs), which, in principle, employ more resource-efficient tools to address the same problem.

Our contribution can be summarised as follows:

- We investigate six different LLM-based approaches to patient-trial matching, comparing them with other neural and statistical models in a pipeline-based setting;
- We propose a fine-tuned LLM approach for further re-ranking of CTs;
- We present an analysis of the computational efficiency of our methods.

We propose a LLM-based method for patient-trial matching that does not rely on task-specific feedback from human experts and yields effectiveness comparable to top systems evaluated in TREC 2023 Clinical Trials. More generally, our work provides a perspective on the current capabilities of LLMs in biomedical information retrieval and provides insights into their applicability concerning the availability of training resources, robustness to domain shift, and efficiency trade-offs.

1.1. Statement of significance

Problem: Matching patients to clinical trials is a complex process that could benefit greatly from the use of advanced neural ranking models.

What is already known: Use of neural re-rankers, including LLMs, shows promising results across a range of information retrieval tasks, including matching patients to clinical trials.

What This Paper Adds: This study introduces methods to leverage LLMs at different stages of the patient-trial matching pipeline. We propose methods that are evaluated in a strict experimental setup with a test set used only once (for multiple systems, as part of official TREC 2023 Clinical Trials evaluations). We analyse the cost-effectiveness of our approaches and compare them against a backdrop of proven neural rankers for patient-trial matching. We also compare the effectiveness of a fine-tuned ranker with established prompting strategies and evaluate the impact of using a more complex LLM.

2. Background

Relevant literature can be divided into two categories: clinical trial search and neural re-ranking.

2.1. Clinical trial search

The success of clinical trials depends on finding subjects who match complex inclusion and exclusion criteria [15,16]. Since 2017, searching over clinical trials has been the focus of Precision Medicine and Clinical Trials tracks in TREC annual challenges [6,7,17–21]. These tracks provided opportunities for developing various search methods of matching patients to clinical trials, such as multi-stage neural ranking system [11], Trial2Vec [22], BERT-based re-ranking methods [8,9,23] and search prototypes such as Science2Cure [24]. Another approach focused on improving the matching process using various information extraction methods [9,25]. Notably, TREC tracks provide benchmarking and evaluation resources that can be reliably reused to assess the effectiveness of new approaches. Here, we leverage these existing TREC Clinical Trials 2021–23 collections [7,20,21].

2.2. Neural re-ranking

Deep learning models in search have shown potential in the information retrieval community as well as in the clinical space [26,27]. Earlier deep re-ranking models [28,29] that used CNNs (Convolutional Neural Networks), however, were unable to adequately capture semantics which is needed for the disambiguation of clinical terminology comprehension [30].

Since then, Transformer [31] models have been introduced with higher capabilities of capturing semantics. Initially, Transformers were used without ranking objectives (so merely as a text encoder). This is largely due to early Transformer-based models, such as BERT [32], being introduced for representation of context with training focused primarily on language understanding tasks like GLUE [33]. Thus early adoption of BERT involved incorporating a BERT-based encoder to re-weight query terms for BM25 [34] or to provide contextualised text representations for downstream processing with PACRR/DRMM [35]. Direct application of BERT as a point-wise ranker was proposed to estimate query-passage relevance [36,37] using BERT's *next sentence prediction* set-up, where both query and the document are fed simultaneously into the model. These point-wise methods can be considered early-interaction models or cross-encoders which we use in this work. Due to high computational intensity these models are usually paired with a computationally cheaper ranking method to perform the initial retrieval (e.g., BM25 or other lexical methods). Alternatively, late interaction models, or bi-encoders, with emphasis on efficiency, appear later, e.g., ColBERT [38]. Most of these models had relatively low-quality embeddings [39], which led to the popularity of contrastive learning for robust embedding representation learning [40,41]. Interaction models, however, have a problem where the pretraining task is not aligned with the ranking task, known as the cold-start problem [42] and may result in worse effectiveness than BM25 for biomedical tasks [43].

2.3. LLMs in clinical patient-trial matching

Existing studies on using LLMs in patient-trial matching largely focus on a setup, where admission criteria are semi-structured and patient notes are evaluated against individual inclusion and exclusion criteria (a 'structure then match' approach). This process is often iterative. Examples of methods addressed in existing studies include aggregation of criterion-level predictions [44], and fine-grained criterion-level classification evaluated on historical admission data in oncology [45]. Wornow et al. [46] evaluate the zero-shot effectiveness of LLMs in a setup closely matching current recruitment protocols, that is, from a perspective of recruitment to a single clinical trial.

In contrast, our methods address patient-trial eligibility assessment within a single LLM call. We also investigate the impact of fine-tuning on the effectiveness of the matching process (against the backdrop of established prompt engineering strategy), and provide insight on domain shift consistent with the setup of the TREC CT 2023 shared task.

3. Data

We conduct our experiments on the TREC Clinical Trials 2021–23 track collections. We use the TREC 2023 Clinical Trials collection (CT 2023 henceforth) for evaluations. The CT 2023 dataset consists of 40 topics¹ with 34,931 relevance judgments corresponding to 37 of the topics (with no evaluation data provided for the remaining 3 topics). The corpus for the task is a May 2023 snapshot of Clinical-Trials.gov database,² with over 450K registered clinical trial records.

¹ Topic is a formatted representation of an information need. In the datasets used here, each information need (topic) is some form of patient data.

² <https://clinicaltrials.gov/>

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<topic number="1" template="glaucoma">
  <field name="primary condition">glaucoma</field>
  <field name="definitive diagnosis">primary open angle glaucoma</field>
  <field name="visual field">moderate field damage</field>
  <field name="visual acuity">0.3</field>
  <field name="prior cataract surgery">no</field>
  <field name="prior LASIK surgery">no</field>
  <field name="comorbid ocular diseases">corneal edema</field>
</topic>

```

Fig. 1. An example topic from the TREC CT 2023 track.

Each topic simulates an *individual* patient's questionnaire on a specific medical condition, therefore the task involves semi-structured topics with several fields (e.g., Fig. 1). Topics are synthetic in a sense that they represent plausible (not actual) patients, and were curated with the involvement of medical professionals.³ Of note, each topic (so, each questionnaire) represents a single patient. This 1:1 relationship between topics and patients also holds for previous TREC Clinical Trials collections discussed further in this section.

Clinical trials are *fielded documents* with the following fields: brief summary, brief title, identifier, detailed description, drug name, drug keywords, inclusion/exclusion criteria, gender, general keywords, intervention type, maximum age, minimum age, official title, and primary outcome. Intervention type, gender, and primary outcome refer to controlled vocabularies; age-related fields are numeric. All other fields except clinical trial ID are textual.

For each evaluated topic-document pair in the dataset, a relevance judgment assigns a score of 0 for *not relevant*, 1 for *excluded* (trials which are topically relevant, but the patient would be excluded), and 2 for *eligible* (trials which are topically relevant, and the patient matches the eligibility criteria).

For a learning-to-rank model tuned on human judgments, we use TREC 2021 Clinical Trials collection (CT 2021) [7] as training data. CT 2021 contains 75 topics and 35,832 corresponding relevance judgments with the same three relevance classes as those of CT 2023. The CT 2021 benchmark is defined for an earlier (2021) snapshot of the same clinical trials database and contains over 375 K trials. A 2022 edition of the track (CT 2022) re-used the same clinical trial corpus with 50 new topics and 35,394 relevance judgements. We use the CT 2022 dataset as a validation set in our experiments.

CT 2021 and CT 2022 using an earlier snapshot of the same CT database as TREC CT 2023 means that nearly all of the trials present in CT 2021/2022 collections will also feature in the CT 2023 document corpus. This being said, each year's collection comes with a unique set of topics and, consequently, a unique set of judgments specific to these topics. We use the entire CT 2021 collection for training, the entire CT 2022 for validation experiments, and the entire CT 2023 for testing.

An important point of difference between the training/validation datasets and the CT 2023 dataset is the format of the topics. In the 2021 and 2022 editions of the track, patients were represented with synthetic admission notes, as opposed to questionnaire data (which is used in CT 2023 topics). An example patient from CT 2022 is presented in Fig. 2. Statistics corresponding to the TREC CT datasets can be found in Table 1. This mismatch between train and test data means that our experiments address an additional research question on ranker robustness to a form of domain shift. In practical settings, this could be akin to training a model on a specific type of patient representation, to be then used with different ones (e.g., in a different medical service area that uses a distinct format for patient notes).

³ This means that TREC patient profiles can be submitted to LLM APIs with no concern about patient privacy.

Table 1
Statistics of TREC CT 2021, 2022, and 2023.

	2021	2022	2023
Documents (clinical trials)	375,580	375,580	451,538
Topics (patients)	75	50	37
Avg. topic length (tokens)	133.4	105.9	29.8
Avg. topic length (sentences/fields)	11.2	9.4	8.7
Total judgements	35,832	35,394	34,931
– Eligible (2)	5,570	3,939	11,963
– Excluded (1)	6,019	3,036	11,262
– Not relevant (0)	24,243	28,419	11,706
Unique trials judged	26,162	26,585	17,106

4. Methods

In this section, we describe the methodology implemented in our study. First, we present the methods used for pre-processing clinical trials and patient descriptions. Subsequently, we outline the retrieval approaches used at different stages of the retrieval pipeline. A schematic overview of a multi-stage clinical trial retrieval pipeline is presented in Fig. 3. Within the pipeline, the first step is to represent the patient questionnaire as a search query. We then carry out the initial retrieval to get a ranked list of 1000 clinical trials and then re-rank the top-k results of this list using (one or two) re-rankers based on language models. The main motivation for using the 'retrieve and rerank' approach is to avoid running the costly re-ranking models on the entire corpus. Hence, we retrieve a pool of plausible candidates with a computationally cheaper model (e.g., BM25 used here, or models evaluated in Appendix B) and re-score the most promising ones (using language models) without running into prohibitive computation costs (we analyse this trade-off further in this paper). The pipeline setup used in this work also allows for an ablation style analysis, hence providing an insight on the added value of re-ranking in a reasonably realistic setting.

4.1. Pre-processing

Query formulation. Topics for the TREC CT 2023 are given in the format of disease "templates" with XML-structured key information. As mentioned in the previous section, compared to previous editions of the track, the topics differ from the format followed in the 2021 and 2022 editions, which considered free-text admission notes instead. We use two alternative query formulation strategies:

- **template** – we simply concatenate XML field names and textual values of XML nodes;
- **free-text** – we reformulate queries such that templates approximate free text admission notes; for that, we formulate a prompt to *gpt-3.5-turbo* model and use the output as queries.

Topic 8: A 7-month-old boy is brought to emergency by his parents due to irritability and inability to defecate for the past 3 days. The patient has had constipation and discomfort with bowel movements since birth. His symptoms worsened after eating semi-solid foods. Vital signs are normal. Abdominal examination shows distension and tenderness to palpation with presence of bowel sounds. Xray with barium shows a narrow rectum and rectosigmoid area. The rest of the colon proximal to this segment is dilated. Digital rectal exam revealed burst of feces out of the anus. The biopsy showed absence of submucosal ganglia in the last segment of the large intestine.

Fig. 2. An example topic from the TREC CT 2022 track.

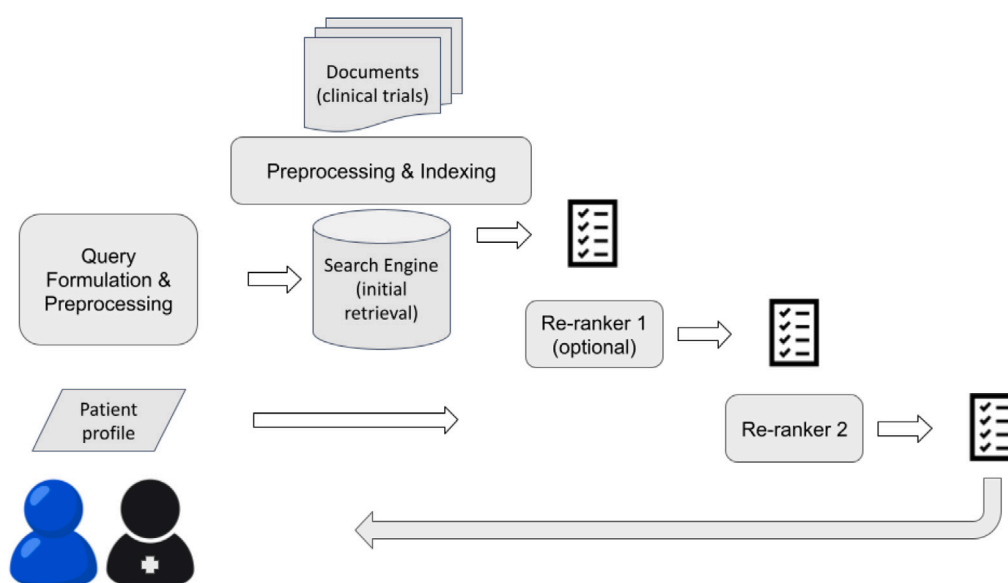


Fig. 3. An overview of a cascading patient-trial matching system implemented as a multi-step retrieval pipeline. After formulating a search query from the patient profile we retrieve an initial set of documents with a computationally cheap model (e.g., BM25) and then re-rank the top portion of this initial ranking (with one or two rankers based on language models).

The free-text strategy is geared towards making the topics similar to those of TREC CT 2021/22 (namely, the training and validation data). To this end, we provide the *gpt-3.5-turbo* model with one example from the CT 2022 collection and explicitly instruct it to keep the note concise. Furthermore, the prompt specifies that the model should not add the patient's age or gender if these details are not present in the XML format. The template strategy is a naive baseline. For trained models using the template strategy, the assumption is that a trained component can work reasonably well on out-of-domain training data.

Query and document enrichment. To implement query and document enrichment we follow the procedure described by Kusa et al. [9,47]. Initially, the content of clinical trials is parsed to identify and separate sections. A heuristic-based parser⁴ is employed to divide the eligibility criteria section into two lists: one containing sentences with inclusion criteria and the other with exclusion criteria. We use two fine-tuned BioBERT-based models [48] to identify disease⁵ and chemical

(i.e., drug)⁶ entities within these criteria. Subsequently, we apply a BERT-based negation detection model⁷ on the previously identified entities. In the final stage, we combine the entities extracted from the exclusion and inclusion criteria sections. This involves inverting the negation modifiers in the exclusion entities (for example, converting the exclusion criterion “has allergies” to the inclusion criterion “has no allergies”). Due to imperfections in the extraction process, where the model sometimes extracts both affirmative and negative versions of the same entity, we give precedence to positive entities for inclusion criteria and eliminate the others as necessary. If an entity consists of several tokens, we concatenate them to create a single token for each entity to preserve the semantic information of each entity. These entities are then used to enrich the documents for lexical retrieval models. An analogous approach is used to pre-process patient descriptions by extracting and normalising disease and drug entities. Furthermore, we use LLM-supported synonym enrichment by prompting the *gpt-3.5-turbo* model to provide normalised synonyms for extracted terms in the

⁴ <https://github.com/WojciechKusa/clinical-trials>

⁵ https://huggingface.co/alvaroaalon2/biobert_diseases_ner

⁶ https://huggingface.co/alvaroaalon2/biobert_chemical_ner

⁷ <https://huggingface.co/bvanaken/clinical-assertion-negation-bert>

query. We then augment the input query with extracted keywords. The method is intended to capture and represent the entities as they appear in the document corpus, along with their negations. We normalise the entities appearing in the queries using the GPT-3.5 model. We apply this method of query and document enrichment exclusively to lexical retrieval (i.e., BM25).

LLM-based query expansion: We propose a query reformulation method based on GPT-3.5-turbo prompting. We assume that re-wording the key information regarding a specific diagnosis can produce useful expansion terms for a given input topic. That is, we record the first paragraph of LLM-generated text for the prompt: ‘<TOPIC TEXT> What is the diagnosis for this patient?’. Importantly, the objective of this expansion method is not to learn the diagnosis from an LLM (the diagnoses are explicitly given in CT 2023 topics) but to prompt the LLM to produce a verbose response that includes valuable expansion terms closely related to the diagnosis. This strategy was originally designed for validation experiments carried out on the CT 2022 dataset, where the diagnosis is not explicitly given for all the topics. However, in validation experiments, we observed that the strategy works (i.e., improves the retrieval effectiveness) regardless of the presence of diagnosis in the topic, hence its application to CT 2023.

4.2. Retrieval models

In this subsection we introduce ranking models used in our experiments. For the initial retrieval step, we use different flavours of well-established BM25 baseline. The use of different implementations gives us insight into the downstream implications of the initial ranking effectiveness.

Our choice of BM25 for initial retrieval is motivated two-fold. Firstly, we focus our experiments on the impact of re-ranking with BERT-based rankers and LLMs, so choosing an efficient and stock-standard baseline appears reasonable. Secondly, BM25 has proved to be an effective approach in many domain-specific retrieval scenarios, such as the one contemplated here. In our experiments, BM25 outperformed dense retrieval models on TREC CT 2021 [49], and it also provided highly competitive results in legal case retrieval [50]. We also report results obtained in post-TREC experiments using dense retrieval approaches (so, potential alternatives to BM25) in Appendix B, which appear to confirm our initial intuition about the strength of BM25 as an initial retriever for CT search.

We also carry out experiments with strong BERT-based re-ranking baselines introduced in our earlier works. The LLM-based ranking methods are novel elements of the retrieval pipeline introduced in this work.

BM25: BM25 is used as an initial retrieval mechanism; we use a Solr [51] search engine’s BM25 implementation. We index clinical trials with all their fields, with inclusion/exclusion criteria split into two separate fields. Moreover, all textual fields are aggregated into a *text* field. For basic BM25 retrieval (denoted BM25), the documents are scored using this aggregated field. The BM25 indexing and retrieval procedure matches the setup described in earlier work by Rybinski et al. (2020) [8]. The baseline run itself can be downloaded from A2A-API [52] at <https://a2a.csiro.au/api/bm25/ct2023>.

Alternatively, we evaluate a BM25 model implemented in ElasticSearch⁸ with the query and document enrichment approach described in the previous section. We refer to this model as BM25-ES in order to distinguish it from the Solr implementation.

SCT-BERT: The SCT-BERT clinical trial re-ranker [23] is fine-tuned using the corpus of the CT 2021 [7] collection in a self-training regime. The model is trained to match *brief summaries* and *inclusion criteria* from the same clinical trial (i.e., to discriminate between a pair of texts representing the same trial and a pair drawn randomly from separate clinical trials). At inference, we use *brief summaries* to represent the document side of the input and topic text to represent the query. The SCT-BERT model is a re-ranker, we calculate the ranking score as a 9:1 interpolation of a softmax over the BERT’s logit output with the ranking score obtained in the previous ranking step (here: BM25), following the setup used in previous work [8,53].

TCRR: The TCRR (Topical and Criteria Re-Ranking) [9] method draws inspiration from curriculum learning principles, where concepts are progressively learned from simple to complex. The model is initially trained on identifying documents of topical relevance, where both eligible and excluded documents are pertinent. Subsequently, it targets the more challenging eligibility classification, distinguishing only eligible documents as relevant [9]. This dual-objective training uses a pre-trained BERT language model as a cross-encoder, enhanced with a linear combination layer for fine-tuning through a pairwise loss function. The inference process mirrors this training by employing the BM25 rank to perform a two-fold re-ranking of top trials, leveraging the distinct models stemming from the dual training objectives.

LLM-based (re-)ranking with embeddings: For embedding-based ranking, we use LLM-based embeddings for pointwise scoring. Let $LLM_e(s)$ denote a LLM-based embedding of text s . To calculate an embedding-based ranking score of document d for a query q , $\rho_{d,q}$, we use:

$$\rho_{d,q} = LLM_e(q) \cdot LLM_e(d_{inc}), \quad (1)$$

where d_{inc} denotes inclusion criteria of clinical trial d . Our final scores are averages of the normalised first stage retrieval (i.e., BM25) score and $\rho_{d,q}$. We refer to this re-ranking approach as GPT_v henceforth.

The re-ranking model is fine-tuned on 10,000 labelled topic-document pairs randomly sampled from TREC CT 2021 relevance judgments. Of note, 2021 topics were in a free-text format simulating an extract from a patient’s electronic health record, so the topics presented to the model at inference are quite different to those used in training.

LLM-based relevance scoring: We obtain prompt-based ranking scores directly by prompting *gpt-3.5-turbo* model (with a prompt ‘Consider the following patient: <TOPIC TEXT>. Is the clinical trial below adequate for this patient: <CLINICAL TRIAL TEXT>. Reply only with a confidence score between 0 and 1.’). The topics are represented in the prompt with the original topic text, while the trials are represented with a concatenation of *brief summaries* and *inclusion/exclusion criteria*. Top k documents are scored directly with the output of the model interpolated with the inverse rank resulting from the previous retrieval step, the remaining documents are scored with the inverse rank from the previous retrieval step. We refer to this re-ranking approach as GPT_r henceforth.

LLM-based filtering: Finally, we also evaluate LLM as a relevance filter in a zero-shot setup using the question-answering approach. We use the following prompt: ‘Clinical trial eligibility criteria section: <ELIGIBILITY CRITERIA>.\n\nPatient description: <TOPIC TEXT>.\n\nIs the patient eligible for the trial? Write only a binary decision (yes/no) as a response. Write ‘no’ if you are unsure. Consider only the criteria concerning diagnosis and current medical condition. Disregard age, gender, geographical location, and other criteria that are not mentioned in the patient’s description.’.

The output from the model is then used to post-process the list of returned clinical trials. Specifically, we exclude clinical trials where the model-generated text starts with ‘no’ (case insensitive). This filtering is

⁸ <https://www.elastic.co/elasticsearch/>

Table 2

A summary of the evaluated runs. The systems are sorted by ascending computational cost. The second last column indicates whether a specific run was evaluated officially within the TREC shared task. QE is short for Query Expansion.

	Formulation	QE	Initial retrieval	1st Re-ranking	2nd Re-ranking	Official	Model name
1	template	–	BM25	–	–	✓	BM25 (baseline)
2	free-text	–	BM25	–	–	–	BM25 (GPT formulation)
3	template	GPT	BM25	–	–	✓	BM25 (GPT expansion)
4	free-text	–	BM25-ES	–	–	✓	BM25 (query & document enrichment)
5	template	GPT	BM25	GPT_v	–	✓	BM25 (GPT exp.)
6	template	GPT	BM25	SCT-BERT	–	–	BM25 (GPT exp.) + SCT-BERT
7	free-text	–	BM25-ES	TCRR	–	✓	BM25 (q&d enr.) + TCRR
8	template	GPT	BM25	GPT_t	–	✓	BM25 (GPT exp.) + GPT_t
9	free-text	–	BM25-ES	TCRR	GPT_F	–	BM25 (q&d enr.) + TCRR + GPT_F
10	free-text	–	BM25-ES	TCRR	GPT_i	–	BM25 (q&d enr.) + TCRR + GPT_i

applied to the top 50 clinical trials returned for each patient from the previous retrieval step, or whenever there were at least ten trials for which the model provided a positive response (whichever condition is met first). We refer to this re-ranking approach as GPT_F henceforth.

5. Experiment setup

We report results on CT 2023. As previously mentioned, all the trained modules are tuned on CT 2021 data. Score fusion methods (score interpolations, rank interpolations) were validated using CT 2022. We note that in many cases the use of CT 2022 was to merely guide design decisions, as the shift in topic semantics and format would likely limit the effectiveness of end-to-end hyperparameter optimisation.

We evaluate ten systems representing different combinations of retrieval pipeline elements. The first four consist of only a statistical first-stage retrieval model with various query expansion, enhancement and formulation approaches. The remaining six use neural re-ranking models. We summarise the runs in Table 2. Of note, the models in Table 2 are sorted by ascending computational cost. For example, row 1 only uses a traditional inverted index and a simple statistical model for scoring matching keywords; rows 2 and 3, apart from an equivalent use of a statistical model, use a single LLM-based generation per query formulation and expansion, respectively. In contrast, the model represented in row 10 uses an LLM generation to formulate the query, the statistical model is enhanced with the results of offline processing (while latency implications are minimal, the results of the offline process need to be stored in the index additionally); the results of the initial retrieval step are then re-ranked at $k = 200$ using SCT-BERT and $k=100$ with TCRR, and subsequently re-ranked again at top $l = 100$ using LLM-based methods.

Importantly, our training-validation-test setup, combined with the fact that most of our runs were evaluated officially within the TREC CT 2023 track, ensures we use the test set only *once*. The same is true for our runs that were not evaluated officially through the shared task — we directly combine elements of officially evaluated runs. For example, the application of LLM-based re-ranking (GPT_t) to run 7 results in run 10. To report our main results we do not optimise score or rank fusion, depth of re-ranking, or other parameters.

We also present an additional set of experiments carried out post-TREC. In these experiments, we investigate the impact of using an established *prompt engineering* approach (i.e., 0-shot and 1-shot chain-of-thought prompting, hereafter referred to as *CoT* (*Chain-of-Thought*) *prompting*) [54], as well as the impact of using a more powerful LLM (i.e., GPT4-o). The CoT prompting templates can be found in Appendix A.

5.1. Evaluation

We follow the evaluation procedure from the TREC Clinical Trials track. As the relevance assessment is given in a graded relevance scale (eligible (2), excluded (1), and not relevant (0)), the performance of

the models is measured using normalised discounted cumulative gain (nDCG). We present results as reported by TREC, using nDCG@1000 and nDCG@10. Furthermore, we use two binary measures that consider only eligible trials as relevant: Precision (P@10) and Reciprocal Rank (RR).

The nDCG evaluation aligns with measuring the effectiveness of the systems with a more traditional (broader) notion of relevance (rather than eligibility). We believe that this might be useful for matching patients to clinical trials, for example where very few ‘eligible’ documents are retrieved, and a detailed examination of reasons for exclusion is needed to see if any of these criteria can be overcome (e.g., undertaking a specific treatment as a pre-requisite to enrollment) or when matching the patients to treatments is done not for enrollment purposes, but, for example, to review prospective treatment options.

Notwithstanding, we aim to identify trials the patient is eligible for, and Precision at rank 10 (P@10) provides strict feedback on achieving this goal. We treat unjudged documents as not relevant, ensuring that our results are not biased towards models that retrieve a large number of unjudged documents.

5.2. Implementation

In all BM25 runs, including the initial retrieval, scores are calculated using the default values for the hyperparameters: $b = 0.7$ and $k1 = 1.2$. We use a version of SCT-BERT trained on the entire 2021 corpus for 1 epoch with a batch size of 16. Full details of the training procedure can be found in the original SCT-BERT paper. Based on the performance on the TREC CT 2022 topics we choose the BlueBERT [55] checkpoint⁹ fine-tuned on the TREC CT 2021 collection as the base model for the TCRR method.

All GPT-based experiments are implemented via OpenAI API. For GPT_v we use the *text-embedding-ada-002* model (used as is, i.e., without additional training), all other GPT-based methods use *gpt-3.5-turbo*. The GPT_t model was trained using 10,000 training samples drawn randomly from the CT 2021 relevance judgements, with one epoch of training and batch size of 6.

6. Results and discussion

We first present the evaluation of the TREC CT 2023 collection. Then, we discuss the effectiveness-efficiency trade-off between our approaches.

⁹ https://huggingface.co/bionlp/bluebert_pubmed_uncased_L-12_H-768_A-12

Table 3

Ranking model evaluation on the TREC Clinical Trials 2023 collection. **Bold** values indicate the highest score in our experiment. Underlined values indicate the highest scores in official TREC evaluations. q&d enr. represents Query and document enrichment.

ID	Model	nDCG@1000	nDCG@10	P@10	RR
1	BM25 (baseline)	0.487	0.619	0.330	0.563
2	BM25 (GPT formulation)	0.476	0.590	0.322	0.529
3	BM25 (GPT expansion)	0.496	0.641	0.354	0.535
4	BM25 (query & document enrichment)	0.364	0.624	0.476	0.627
5	BM25 (GPT exp.) + GPT_v	0.497	0.652	0.368	0.483
6	BM25 (GPT exp.) + SCT-BERT	0.496	0.657	0.389	0.522
7	BM25 (q&d enr.) + TCRR	0.370	0.684	0.584	0.642
8	BM25 (GPT exp.) + GPT_l	0.532	0.738	0.527	0.667
9	BM25 (q&d enr.) + TCRR + GPT_F	0.366	0.720	0.592	0.735
10	BM25 (q&d enr.) + TCRR + GPT_l	0.375	0.777	0.697	0.783
	Official TREC median	–	0.648	0.397	0.538
	System A	0.400	<u>0.811</u>	<u>0.735</u>	<u>0.785</u>
	System B [56]	0.299	0.638	0.627	–
	System C [57]	<u>0.532</u>	0.738	0.527	0.667
	System D [58]	0.487	0.639	0.362	0.519

Table 4

Latency of individual components of the information retrieval pipeline. The system latency column presents the mean query latency of a hypothetical system, which uses BM25, QE, and one of the rerankers (pipeline given in parenthesis).

Method	Component latency	System latency
BM25	0.15 ± 0.01	0.15 (BM25)
GPT-based QE	1.42 ± 0.81	1.57 (BM25+GPT exp.)
SCT-BERT@200	1.69 ± 0.17	3.26 (BM25+GPT exp.+SCT-BERT@200)
GPT_F @100	45.09 ± 1.29	46.51 (BM25+GPT exp.+ GPT_F @100)
GPT_l @100	46.48 ± 1.54	47.89 (BM25+GPT exp.+ GPT_l @100)

6.1. Most effective approaches on CT 2023

A comparison of different ranking strategies is presented in Table 3. We attain the best results with systems that leverage re-ranking with a trained LLM, which highlights the effectiveness of the proposed strategy to use the model's graded output directly in a re-ranking process via rank fusion (see rows 8 and 10 of Table 3). These results indicate that a trained LLM model (i.e., the GPT_l re-ranker), albeit costly, has a good ability to discern between eligible, not eligible, and not relevant trials for a given patient (see rows 9 and 10 for a direct comparison of a zero-shot and a fine-tuned re-ranker, respectively). At the same time, we observe that re-ranking with an LLM in a zero-shot setup does not yield substantial improvement over the PLM-re-ranked input (row 9 vs row 7). It is worth noting that row 9 represents our best setup leveraging zero-shot LLM re-ranking among those reviewed in preliminary experiments. This seems to indicate that in scenarios where the elevated cost of per-document LLM calculations is not prohibitive, it is worth investing in model fine-tuning.

Compared to official TREC CT results from other systems, our LLM-re-ranked runs situate well above the TREC median. The run presented in row 8 (BM25 with GPT query expansion and GPT_l re-ranking) attained the highest nDCG score across all officially evaluated systems (System C in Table 3 is, in fact, the run from row 8). Our run 10 (BM25-ES with TCRR and LLM re-ranking) would rank second in P@10, only behind a system trained on an in-domain training dataset curated specifically for the shared task [21] (System A in Table 3). This demonstrates that our LLM re-ranking approach is on par with top systems officially evaluated in TREC CT 2023.

Our experiments with LLM tuning show that GPT requires a substantial amount of training data to lead to improvements. In validation experiments on CT 2022, GPT_l did not improve over the previous ranking step when tuning was performed on 1000 or 2000 human judgements, with minor improvements shown with training on 3000 samples. Due to the cost of the tuning process, we tuned the model on 10 K training samples, which yielded the best results on the validation set. We have not explored prospective model improvements with

larger training data volume, but this finding contrasts with conclusions of previous research on smaller PLMs that converge quickly on text tasks [11,23]. On a positive note, the tuned model coped well with the difference between the topic format in the training data (CT 2021) and the test data (CT 2023). Our results appear to indicate that a fine-tuned LLM is robust to the domain shift, which could facilitate the sourcing of training data in some practical scenarios.

We note that the rank fusion approach used in GPT_l re-ranking means that the previous ranking step effectively works as a tie-breaker at every label generated by the LLM re-ranker. This means that the results of GPT_l rely heavily on the effectiveness of the previous step. This could be alleviated by using (pseudo)probabilities over the three labels instead, thus producing a continuous ranking with *gpt-3.5-turbo*; however, this feature is not supported by OpenAI API at the time of writing this manuscript.

In BM25-based methods, we observe the best results with BM25-ES for P@10 and with BM25 with GPT expansion for overall nDCG. Comparison between different BM25 flavours, especially the relatively poor effectiveness of GPT-formulated query with plain BM25 highlights the benefits of the document and query enrichment process in BM25-ES. As mentioned previously, we also carry out a post-TREC experiment to quantify the effectiveness of our BM25 baseline systems against modern dense retrieval approaches. It turns out that BM25 provides a strong initial retrieval baseline for TREC CT 2023. The results of the experiments are presented in Appendix B.

Finally, PLM-based re-ranking and query or document expansion and enrichment, while behind LLM-based re-ranking, appear to provide more efficient alternatives. While the effectiveness gap between, for example, rows 6 and 8 is substantial, we note that SCT-BERT is a zero-shot method. Additionally, the latency induced by the use of LLMs for pointwise re-ranking has to be taken into account in practical setup, which is explored in the next section.

6.2. Effectiveness-efficiency trade-off

We measure query latency under identical conditions from a representative sample of our runs: a BM25 baseline (row 1), BM25 with GPT-based query expansion (row 3), BM25 with GPT QE and SCT-BERT re-ranking (row 6), and BM25 with GPT QE and GPT_l re-ranking (row 8).

Runs 2–4 will have a similar execution time. Run 7 can have a similar execution time to run 6, but will require double the GPU memory for an equal value of the k parameter in top- k re-ranking and a parallel implementation; double time overhead is the worst-case scenario with in-line implementation.

The comparison of single query execution times is presented in Fig. 4. The trade-off between the cost and effectiveness of LLM-based

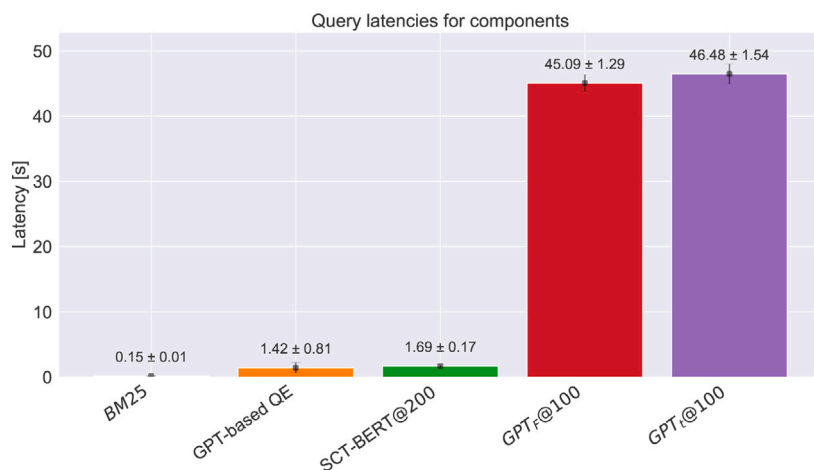


Fig. 4. Comparison of query latencies for a representative subset of methods.

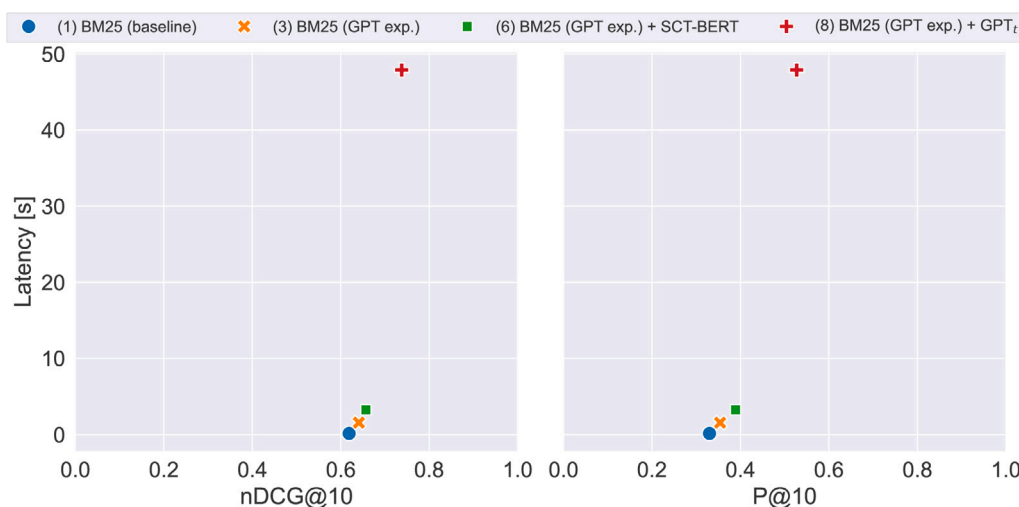


Fig. 5. Comparison of query latencies against effectiveness gains (nDCG@10 and P@10).

re-ranking is illustrated in Fig. 5. We present the latencies of individual pipeline components in Table 4.

We can see that almost 15x larger query latency for the usage of GPT_r model converts to an increase of 12% in nDCG@10.

Our results show that, although the effectiveness improvement is substantial with LLM re-ranking, the associated overhead would need to be carefully considered in practical applications. In addition to elevated query latencies, LLMs induce considerable infrastructure costs. In our experiments, we re-ranked 100 documents per synthetic patient profile using the OpenAI API to a fine-tuned *gpt-3.5-turbo*. API usage cost amounted to around 0.25 USD per 100 API calls (so to rank trials for a single patient).¹⁰ While *running* costs of an LLM-based system could be reduced by a local deployment (which would also be desirable in healthcare settings due to patient privacy protection obligations), such an arrangement would incur additional infrastructure costs, possibly also personnel costs. On the other hand, we note that LLM-associated query latencies and operational costs will most likely decrease in the future.

6.3. Prompt engineering experiments

In our CoT prompting experiments we use a strategy similar to the one used in our fine-tuning approach and our original zero-shot LLM

filtering approach: we re-score each document at the top 100 ranks by interpolating the output of the LLM with an inverse rank of the document in the input ranking with a high smoothing factor. We keep the input ranking for the documents at ranks below 100. In practice, this means that we re-rank top documents by LLM-generated scores while preserving ranking order from the input ranker in case of tied LLM scores. This strategy consistently improves results when compared to scoring with raw LLM outputs, across all the experiments. The results of CoT prompting with *gpt-3.5-turbo* are contrasted with relevant results from our main experiment in Table 5.

Our key findings from the re-ranking with CoT prompting of the GPT-3.5-turbo model can be summarised as follows:

- Re-ranking with CoT does not seem to improve effectiveness much for most evaluation metrics, with the exception of reciprocal rank when compared to input rankings.
- Re-ranking with CoT does not seem to improve over our previous zero-shot re-ranking strategy, *GPT_F*.
- Re-ranking with CoT yields notably lower effectiveness than that of reranking with a fine-tuned model.

We also note, that while CoT is cheaper at inference than a fine-tuned model (in terms of the actual cost of API calls), CoT prompting results markedly longer responses, leading to a higher query latency (average of re-ranking time of ca. 88 s compared to ca. 49 s average

¹⁰ As of March 2024.

Table 5

Results of zero-shot and 1-shot reranking with CoT prompting.

ID	Model	nDCG@1000	nDCG@10	P@10	RR
	BM25 (GPT exp.) w. CoT 0-shot	0.497	0.659	0.392	0.642
	BM25 (GPT exp.) w. CoT 1-shot	0.496	0.654	0.408	0.562
	BM25 (q&d enr.) + TCRR w. CoT 0-shot	0.368	0.652	0.541	0.718
	BM25 (q&d enr.) + TCRR w. CoT 1-shot	0.368	0.698	0.600	0.690
3	BM25 (GPT expansion)	0.496	0.641	0.354	0.535
7	BM25 (q&d enr.) + TCRR	0.370	0.684	0.584	0.642
8	BM25 (GPT exp.) + GPT_i	0.532	0.738	0.527	0.667
9	BM25 (q&d enr.) + TCRR + GPT_F	0.366	0.720	0.592	0.735
10	BM25 (q&d enr.) + TCRR + GPT_i	0.375	0.777	0.697	0.783

Table 6

Results of zero-shot and 1-shot reranking with CoT prompting.

ID	Model	nDCG@1000	nDCG@10	P@10	RR
	BM25 (GPT exp.) w. CoT (GPT4-o)	0.504	0.785	0.603	0.844
	BM25 (q&d enr.) + TCRR w. CoT (GPT4-o)	0.374	0.760	0.692	0.838
	BM25 (q&d enr.) + TCRR w. GPT_4_F	0.372	0.750	0.708	0.797
8	BM25 (GPT exp.) + GPT_i	0.532	0.738	0.527	0.667
10	BM25 (q&d enr.) + TCRR + GPT_i	0.375	0.777	0.697	0.783

of the fine-tuned GPT_i model). The increased time cost of generating longer texts is further exacerbated for larger LLMs (see the next subsection).

One advantage of CoT is that CoT responses can be scrutinised by human actors to assess the reasoning presented by the LLM, while our original method acts largely as a black box. However, we note that this flavour of explanations can also be generated post-hoc on demand, if a human interaction with the search results requires it.

6.4. GPT4-o experiments

We also carry out experiments with a larger (and newer) LLM ranker – GPT4-o (which was not available at the time of our submission to TREC 2023). Again, we report the results of re-ranking runs 3 and 7. For the re-ranking of run 3, we use zero-shot CoT prompting, and for run 7 we use both zero-shot CoT prompting and our original zero-shot prompting strategy (so, an identical prompt to that of GPT_F), here denoted as GPT_{4_F} . We use an identical interpolation strategy to that of CoT prompting experiments. The results are presented in Table 6

Our experiments show that GPT4-o achieves zero-shot ranking effectiveness comparable to that of fine-tuned *gpt-3.5-turbo*, even surpassing the fine-tuned model on some of the effectiveness metrics. Considerable increase in reciprocal rank combined with a lack of improvement on nDCG@1000 might suggest that GPT4 is precise (at the expense of recall) when compared to other ranking methods analysed in our work. The efficiency – effectiveness considerations are an important factor with GPT4 – while CoT prompting appears to yield minimal effectiveness improvements with GPT4-o, it is significantly slower (with medium ranking latency of ca. 242 s, compared to ca. 61 s with GPT_{4_F}).

6.5. When to use LLMs and how?

To highlight the main conclusions from the presented experiments we present the following summary of key findings and their practical implications:

- Largest effectiveness gains are achieved through trained-LLM-based re-ranking; this effect can be observed by comparing rows 7 and 10, and rows 3 and 8 in Table 3; this set-up is also the most costly among those compared in our main experiments, as indicated in Fig. 5.
- In cases where the high latencies of an LLM-based re-ranking are acceptable, it is beneficial to opt for a trained model; compare rows 9 (zero-shot) and 10 (fine-tuned).

- A larger model (here GPT4-o) can provide a competitive zero-shot alternative to a fine-tuned gpt-3.5-turbo model; it will, however, be slower and more expensive.
- PLM re-ranking is a good cost-effective alternative to LLM re-ranking; see rows 4–7–10 for a relevant ‘ablation’ experiment – re-ranking with a simple model (row 7) gets 0.108 improvement over corresponding baseline (row 4) compared to 0.221 improvement with the full pipeline (row 10); according to our latency estimates system from row 7 will also be more than five times faster; notably, the green square representing a PLM re-ranking in Fig. 5 (which covers an analogous ablation experiment with rows 1–3–6–8) shows Pareto-efficiency of the approach; however, effective implementation of such system does rely on the availability of GPU resources, so throughput has to be considered.

7. Conclusions

We investigated the use of Large Language Models (LLMs), in particular GPT models, in matching patients to clinical trials, focusing on their integration into various stages of a retrieval pipeline. We found that LLMs, particularly when fine-tuned, significantly improved the effectiveness of the patient-trial matching process, outperforming other methods in key metrics such as nDCG and Precision.

However, the superior effectiveness of LLMs comes with higher computational costs, highlighting a trade-off between effectiveness and efficiency. This is crucial in practical applications where resources may be limited. Despite the differences in topic formats between our training and test datasets, the LLMs demonstrated robustness and adaptability, suggesting potential for broader applications in medical information retrieval.

While LLMs hold great promise for improving clinical trial matching, balancing their retrieval effectiveness with computational efficiency remains a key challenge for real-world deployment. Future research should aim to optimise this balance and explore the wider applicability of LLMs in healthcare information retrieval.

CRedit authorship contribution statement

Maciej Rybinski: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Wojciech Kusa:** Writing – review & editing, Writing – original draft, Software, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Sarvnaz Karimi:** Writing

– review & editing, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Allan Hanbury:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. CoT prompt templates

A.1. Zero-shot prompt

Consider this patient:
QUERY.

Is the following clinical trial relevant to the patient:

DOCUMENT.

Answer by outputting a relevance score between 0 and 1, where 0 means the patient has a different disease than the one targeted by the clinical trial, score of 0.5 means the disease of the patient matches that of the trial, but the patient description clearly does not match some of the inclusion/exclusion criteria. Score of 1 is assigned to patients who are likely eligible to enter the trial. Please explain your reasoning step-by-step and output the final score SEPARATELY in the last line of the answer, with no other words or symbols in this line.

A.2. One-shot prompt

Consider this patient:

The patient is a 20 year old Caucasian female presents to the clinic with one-sided vision lost and facial weakness, dysarthria and numbness lasting for 1 day. She visited her PCP and underwent brain MRI which revealed a single plaque in the brainstem. After few months, she experienced lower extremities weakness led to balance problem. The second MRI revealed another lesion in the left cerebral hemisphere. The diagnosis of Relapsing Remitting Multiple Sclerosis (RRMS) is confirmed after the second MRI. Her past surgical history is significant for a C-section 2 years ago and she has a one child. She is divorced and is not currently in any sexual relationship. She smokes 10 cigarettes per day and drinks alcohol occasionally. She is working as an editor in a publisher company and she is happy that she can keep working from home most of the time. She is under the treatment of RRMS from 7 months ago.

We are going to assess if the patient is eligible to participate in the following clinical trial:

BRIEF SUMMARY

The purpose of this study is to determine the safety and efficacy of natalizumab in the treatment of individuals who have been diagnosed with relapsing remitting multiple sclerosis (MS). It is hoped that natalizumab will prevent certain types of white blood cells from moving out of the bloodstream into organs, including the brain, that are being damaged by autoimmune disease (a disease in which the body's own immune system attacks certain organs). These white blood cells are thought to cause inflammation that can result in lesions (small areas of damage) in the brain. These lesions are thought to be the cause of relapses and disability in MS.

INCLUSION

– Diagnosis of MS, as defined by McDonald et al. criteria # 1--4 (McDonald et al. 2001) – Between the ages of 18 and 50, inclusive. – Baseline EDSS score between 0.0 and 5.0, inclusive. – Have experienced at least one relapse within the 12 months prior to randomisation. – Cranial MRI scan demonstrating lesion(s) consistent with MS. – Have given written informed consent to participate in the study.

EXCLUSION

– Primary progressive, secondary progressive, or progressive relapsing MS. – MS relapse has occurred, in the opinion of the investigator, within 50 days prior to randomisation and/or the subject has not stabilised from a previous relapse. – A clinically significant infectious illness within 30 days prior to randomisation. – History of, or abnormal laboratory results indicative of any significant cardiac, endocrinologic, hematologic, hepatic, immunologic, metabolic, urologic, pulmonary, gastrointestinal, dermatologic, psychiatric, renal and/or other major disease, that in the opinion of the investigator, would preclude the administration of a recombinant humanised antibody immunomodulating agent for 116 weeks. – History of severe allergic or anaphylactic reactions or known drug hypersensitivity. – Unable to perform the Timed 25-foot Walk, 9HPT, and PASAT 3. – Abnormal blood tests performed at the Screening Visit.

We will assign a relevance score based on the patient's eligibility. 0 means the patient has a different disease than the one targeted by the clinical trial, score of 0.5 means the disease of the patient matches that of the trial, but the patient description clearly does not match some of the inclusion/exclusion criteria. Score of 1 is assigned to patients who are likely eligible to enter the trial. Let us take it step by step:

1. The disease of the patient matches that of the clinical trial, as the patient suffers from a form of MS and the trial focuses on MS treatment. The patient will have a score of at least 0.5.

2. Regarding the inclusion criteria:

- the patient has diagnosis consistent with MS
- the patient is 20 years old, so between 18 and 50
- her onset (relapse) occurred 7 months ago, she has since then stabilised and is working from home
- the medical note does not mention severe disability, so we can reasonably assume she fits into the EDSS range of 0 to 5
- her onset/relapse was 7 months ago, so less than 12 months prior to prospective enrollment to the trial

Table B.7
Effectiveness of the evaluated dense retrieval models on TREC CT 2023.

ID	Model	nDCG@1000	nDCG@10	P@10	RR
	BM25 (baseline)	0.487	0.619	0.330	0.563
	MedCPT	0.278	0.282	0.205	0.374
	Mistral E5	0.339	0.431	0.278	0.508

- her cranial MRI does show lesions consistent with MS

- we can reasonably assume she can give written consent to join the study

3. Regarding the exclusion criteria:

- The note does not mention progressive MS presentation

- The note does not mention a recent relapse in the last 50 days

- The note does not mention infectious diseases or abnormal lab results

- We can reasonably assume that the patient could complete the required physical test, and the note does not contradict this assumption.

4. We can conclude that the patient, based on the provided note, might be eligible to enroll in the natalizumab trial, or at least their medical note does not explicitly contradict this.

5. Based on the above we assign the final score of: 1

Now, consider this patient:

QUERY.

Is the following clinical trial relevant to the patient:

DOCUMENT.

Answer by outputting a relevance score between 0 and 1, where 0 means the patient has a different disease than the one targeted by the clinical trial, score of 0.5 means the disease of the patient matches that of the trial, but the patient description clearly does not match some of the inclusion/exclusion criteria. Score of 1 is assigned to patients who are likely eligible to enter the trial. Please explain your reasoning step-by-step and output the final score SEPARATELY in the last line of the answer, with no other words or symbols in this line.

Appendix B. Dense retrieval experiments

We carry out post-TREC dense retrieval experiments with two of the well known dense retrieval models (general purpose Mistral E5 [59], and a biomedical retriever, MedCPT [60]) to provide a comparison for the effectiveness of our BM25 baseline. The results are shown in Table B.7.

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